

Quantity One[®] Software

Update training guide for version 4.3.0

Dear Quantity One User

Quantity One is a powerful, flexible software package for imaging and analyzing 1-D electrophoresis gels, dot blots, arrays and colonies. Bio-Rad Laboratories is proud to release yet another version of this cornerstone of The Discovery Series suite of software applications.

This training update bulletin is intended to highlight the new features in Quantity One version 4.3.0. The bulletin is not intended to offer full training on the use of the software, please refer to the User Guide, on-line Help function and CD Tutorial for detailed descriptions. The screenshots shown in this bulletin are derived from the Windows version of the application; however, functions are similar in the Macintosh version.

Many of the new features and enhancements in this version of Quantity One are the direct result of conversations with our customers. Please let us know what you would like to see in the next version of Quantity One by faxing, calling or emailing our Technical Services staff. You can also use the Solobug program (automatically installed with Quantity One) to submit software feature requests and suggestions.

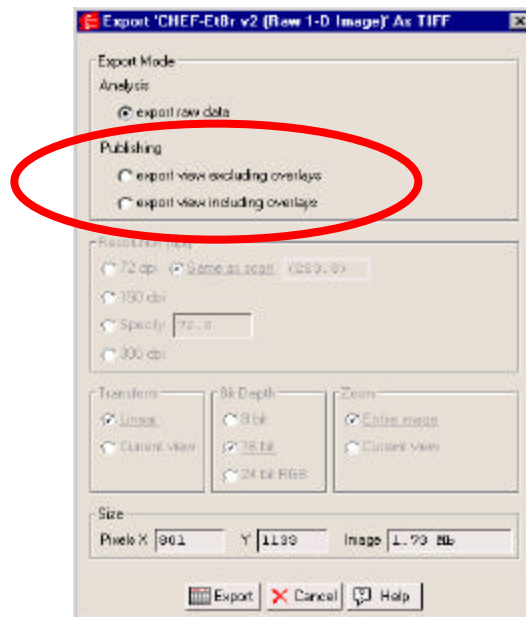
Summary of changes incorporated in Quantity One 4.3.0:

Export an image in TIFF format including overlays	2
Band Type Presence/Absence Report.....	4
Native printing methods for Windows/Macintosh	6
Image Stack Tool allows reordering of images.....	7
Auto Imitate Zoom option is added to Preferences.....	8
Convenient right-mouse-click menu.....	9
Standard report generator for selected reports.....	10
Lane background subtraction dialog improved.....	12
Choice of which imager will be displayed in the file menu	13
Transform window has two added functions	14
Close all function added (only new for Windows).....	15
New file system navigation (only new for Macintosh)	15
Text Tool remembers the last font, color, size and orientation used.....	15
Version history.....	16
Ordering information	17
Network licenses	17
Upgrading from an older version of Quantity One?.....	17
Extended warranty.....	17

Export an image in TIFF format including overlays

Capturing views of analyzed gel data is an important advantage when working with gel analysis software regardless whether the interest is to generate a poster, support a journal paper, or simply to paste into a lab notebook. In previous versions of Quantity One, it was possible to export an image in TIFF format, in two different modes. Analysis mode exports the scan data, unmodified by any viewing adjustments, for analysis in other applications. Publishing mode exports a TIFF image that looks like the image as it is currently displayed on the screen, for publication purposes. In Quantity One 4.3, the capabilities of publication export have been expanded to include the overlays on the image, as it appears on the screen.

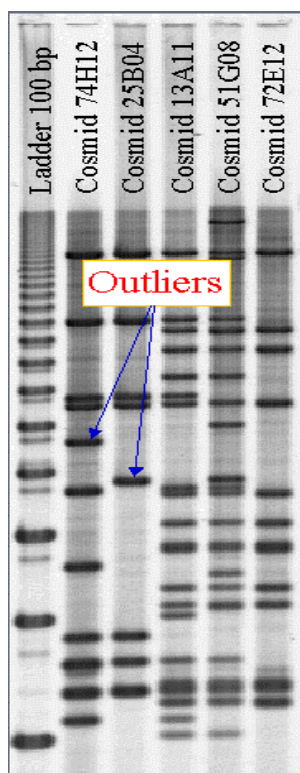
When an image is open and active, File>Export to TIFF Image opens the new dialog box in which you can specify your export parameters:



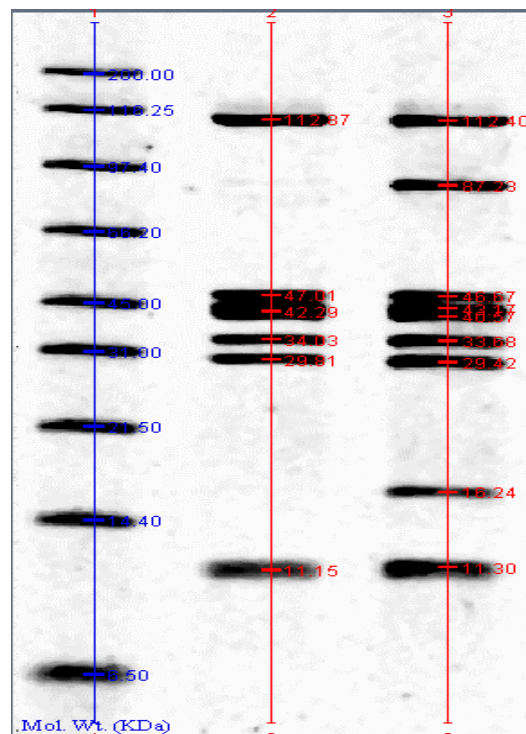
Export>Export to TIFF Image

This new feature enables Quantity One users to analyze and annotate the images within Quantity One and then easily export and open the full view into other documents. The exportable overlays include both manually applied annotations as well as the analytical overlays applied by the software. The exportable overlays include:

- Text annotations
- Volume overlays
- Volume Labels
- Lane and band detection
- Match model lines
- Band labels



Exported annotations



Exported lane and band overlays

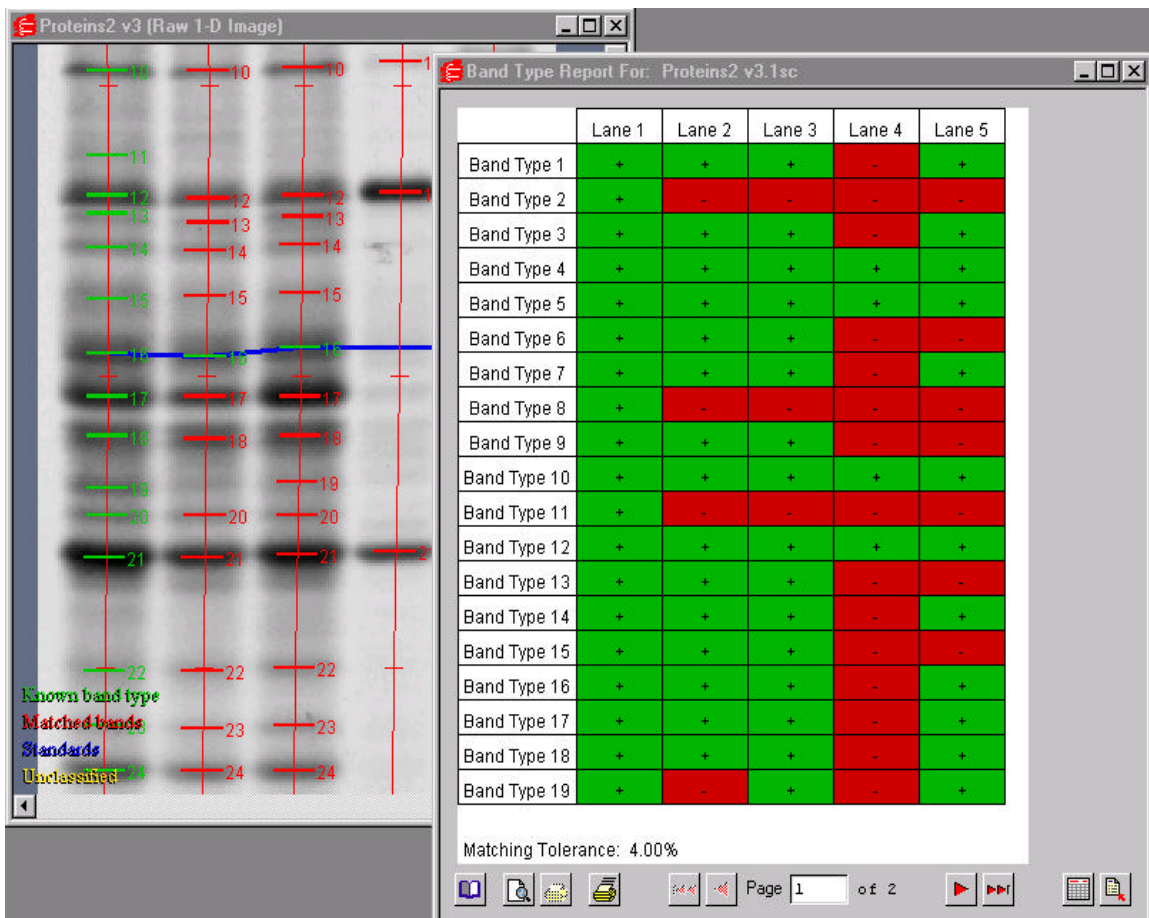
Only the overlays that are currently visible in the image will be exported. The overlays can be hidden with the command View>Hide overlays. The overlays are shown again with the following commands from the menu:

To view	Select
Text annotations and Volume overlays	Edit>Text Overlay Tools
Lane and band detection	View>Show Lanes and Bands
Volume Labels	Volume>Show/Hide Volume Labels

Band Type Presence/Absence Report

It is sometimes desirable to generate a simple qualitative report that indicates the presence or absence of a band across the lanes in a gel. A new report is available once matching has been performed on the image and band types are assigned. The new report displays presence or absence of specific band types across the lanes. The report is selected from Reports>Band Type Report.

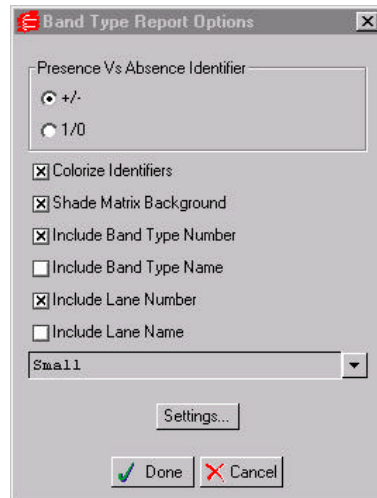
Note! The report can only be selected if the lanes and bands are matched.



Reports>Band Types Report

When the Band Type Presence/Absence Report is selected you have the choice of what to display in the report. The report can be customized for symbols, color and shading identifiers.

Tip! The Lane Name may be edited in Edit>Gel Layout and the Band Type Name may be edited in Match>Match Band Set and then clicking on the lane. The settings may be saved for future use of the report.

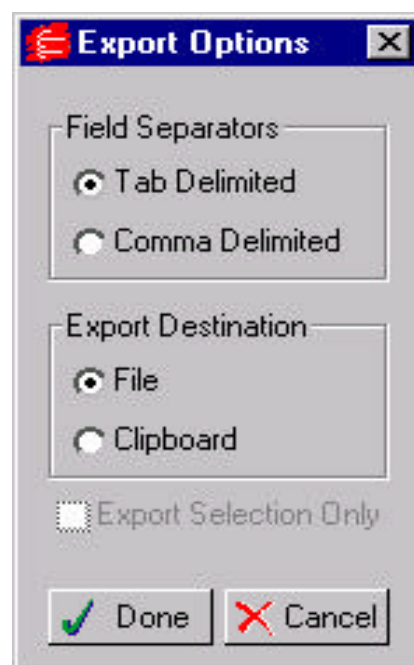


Band Types Report Options

As with other Quantity One reports, the data can easily be exported to other applications, for example, as tab delimited format into Excel.

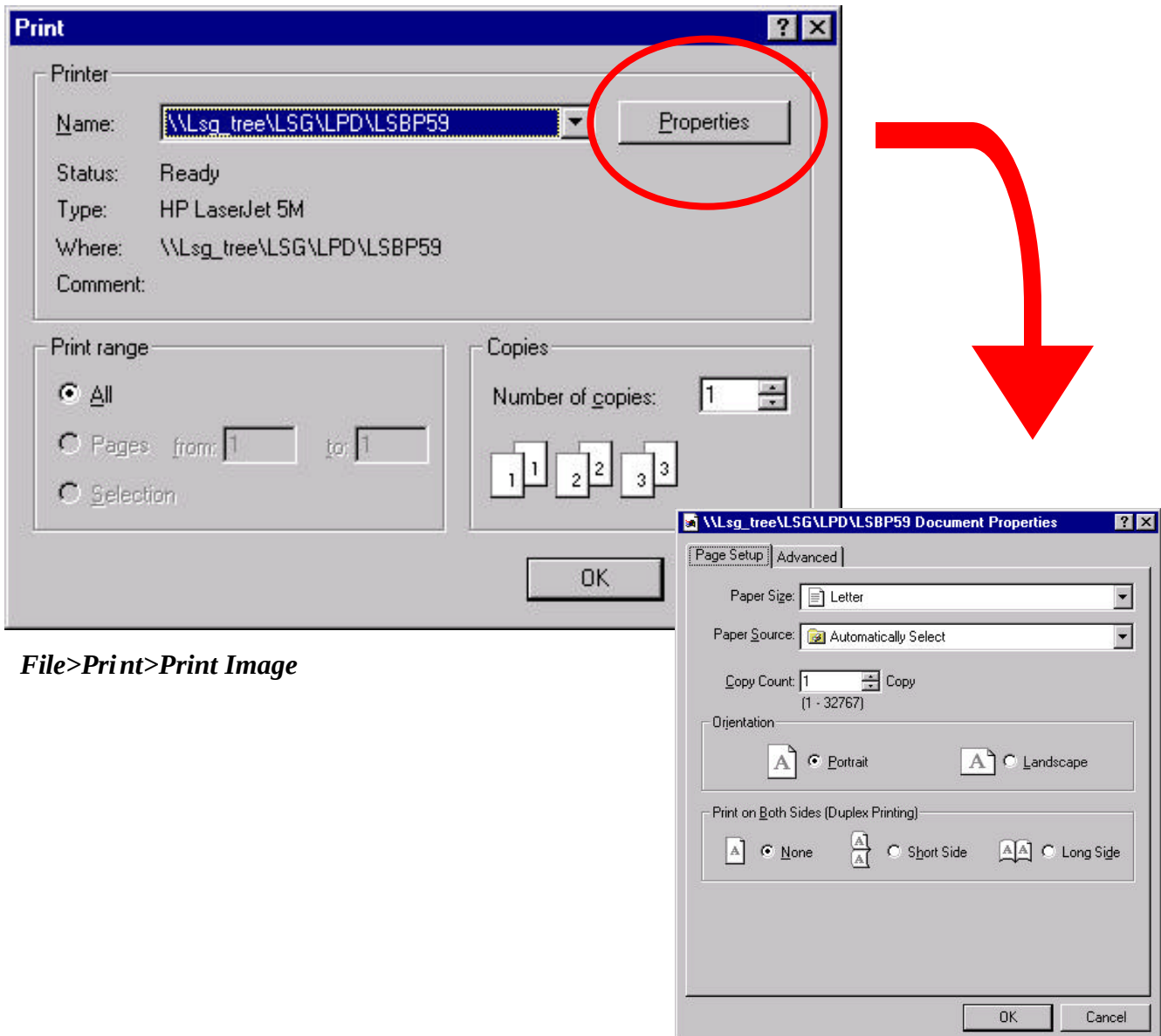
The 'Band Type Report For: Presence? vs Absence?' dialog box displays a matrix of band types across five lanes. The matrix has 19 rows (Band Type 1 to Band Type 19) and 5 columns (Lane 1 to Lane 5). Each cell contains a '+' or '-' symbol, indicating the presence or absence of a band. The background of the matrix is shaded green and red. A red circle highlights the 'Export' button in the bottom right corner of the dialog box.

	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5
	#1	#2	#3	#4	#5
Band Type 1	no name	+	+	+	+
Band Type 2	no name	+	+	+	+
Band Type 3	no name	+	+	+	+
Band Type 4	no name	+	+	+	+
Band Type 5	no name	+	+	+	+
Band Type 6	no name	+	+	+	+
Band Type 7	no name	+	+	+	+
Band Type 8	no name	+	+	+	+
Band Type 9	no name	+	+	+	+
Band Type 10	no name	+	+	+	+
Band Type 11	no name	+	+	+	+
Band Type 12	no name	+	+	+	+
Band Type 13	no name	+	+	+	+
Band Type 14	no name	+	+	+	+
Band Type 15	no name	+	+	+	+
Band Type 16	no name	+	+	+	+
Band Type 17	no name	+	+	+	+
Band Type 18	no name	+	+	+	+
Band Type 19	no name	+	+	+	+



Native printing methods for Windows/Macintosh

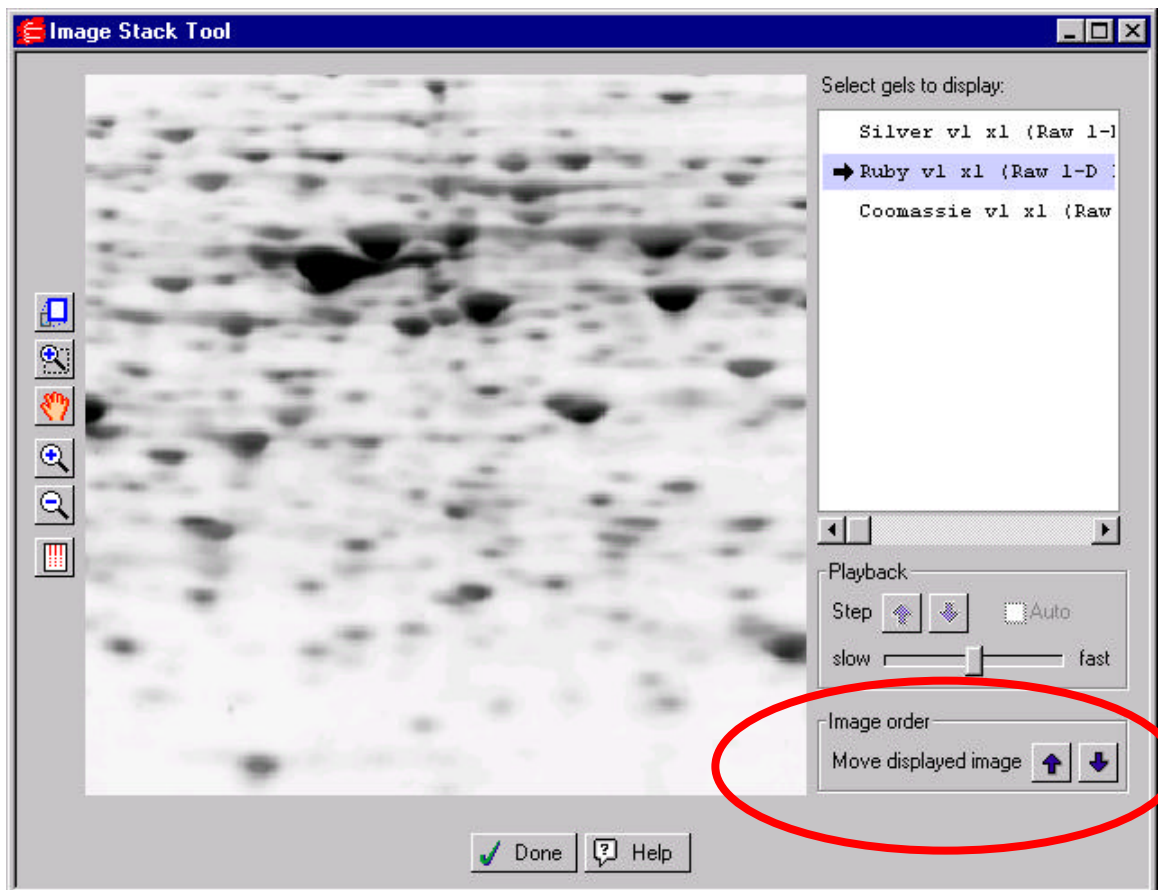
Previous versions of Quantity One have used proprietary printer interfaces. The printer code has been re-written to allow the use of standard Windows and Macintosh interfaces for printing. These native printing interfaces allow the user to print selected pages or to select other document properties. In addition, utilizing native printing methods minimizes compatibility problems with new printers as they are introduced.



File>Print>Print Image

Image Stack Tool allows reordering of images

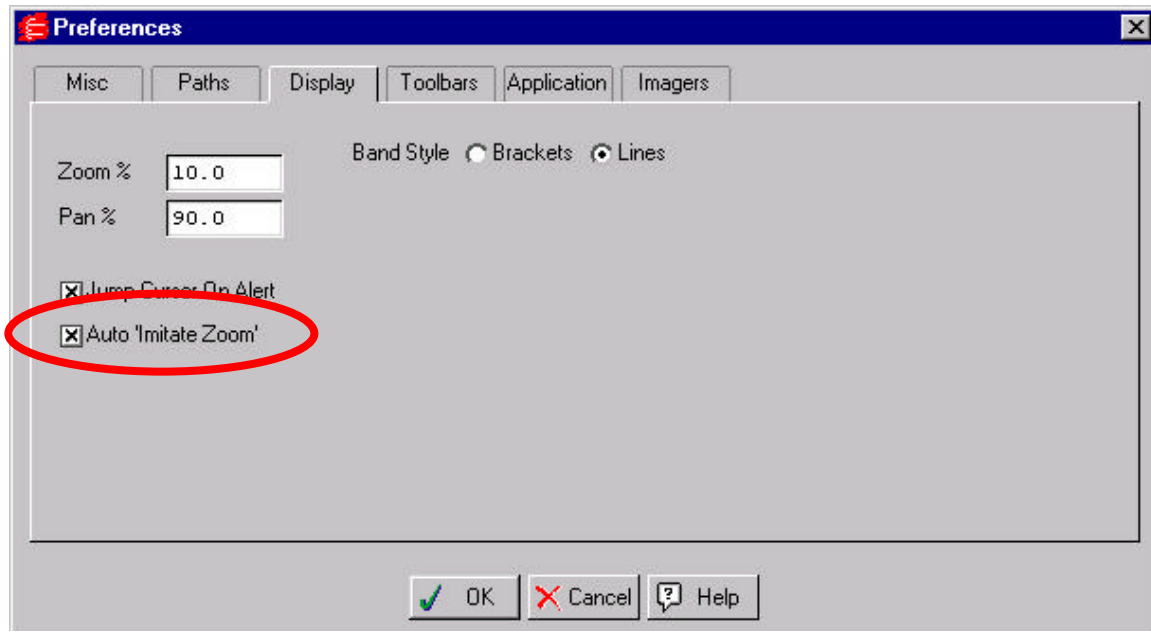
When multiple images are viewed sequentially with the Image Stack Tool, there is now a new button that allows reordering the displayed images conveniently. This is particularly useful for reviewing multiple samples that are related to each other in an ordered manner, for example by time or concentration.



View>Image Stack Tool

Auto Imitate Zoom option is added to Preferences

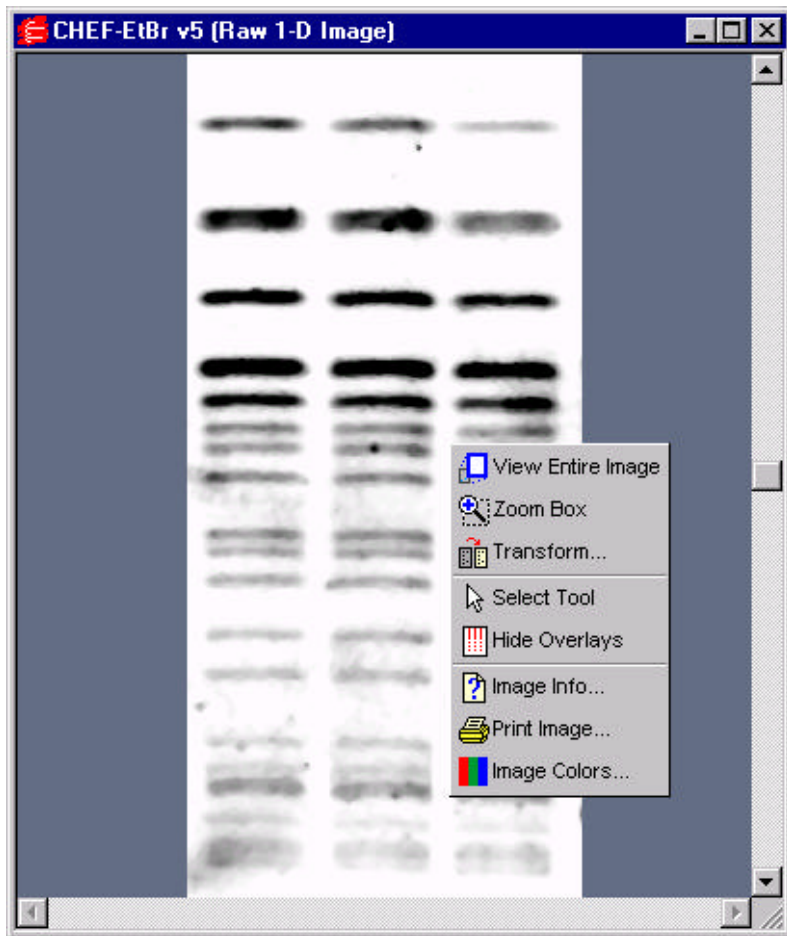
If multiple images are displayed on the screen, they may all be identically zoomed using the new Auto “Imitate Zoom”. The function can be turned off if it is preferred to individually zoom each open image.



Edit>Preferences>Display

Convenient right-mouse-click menu

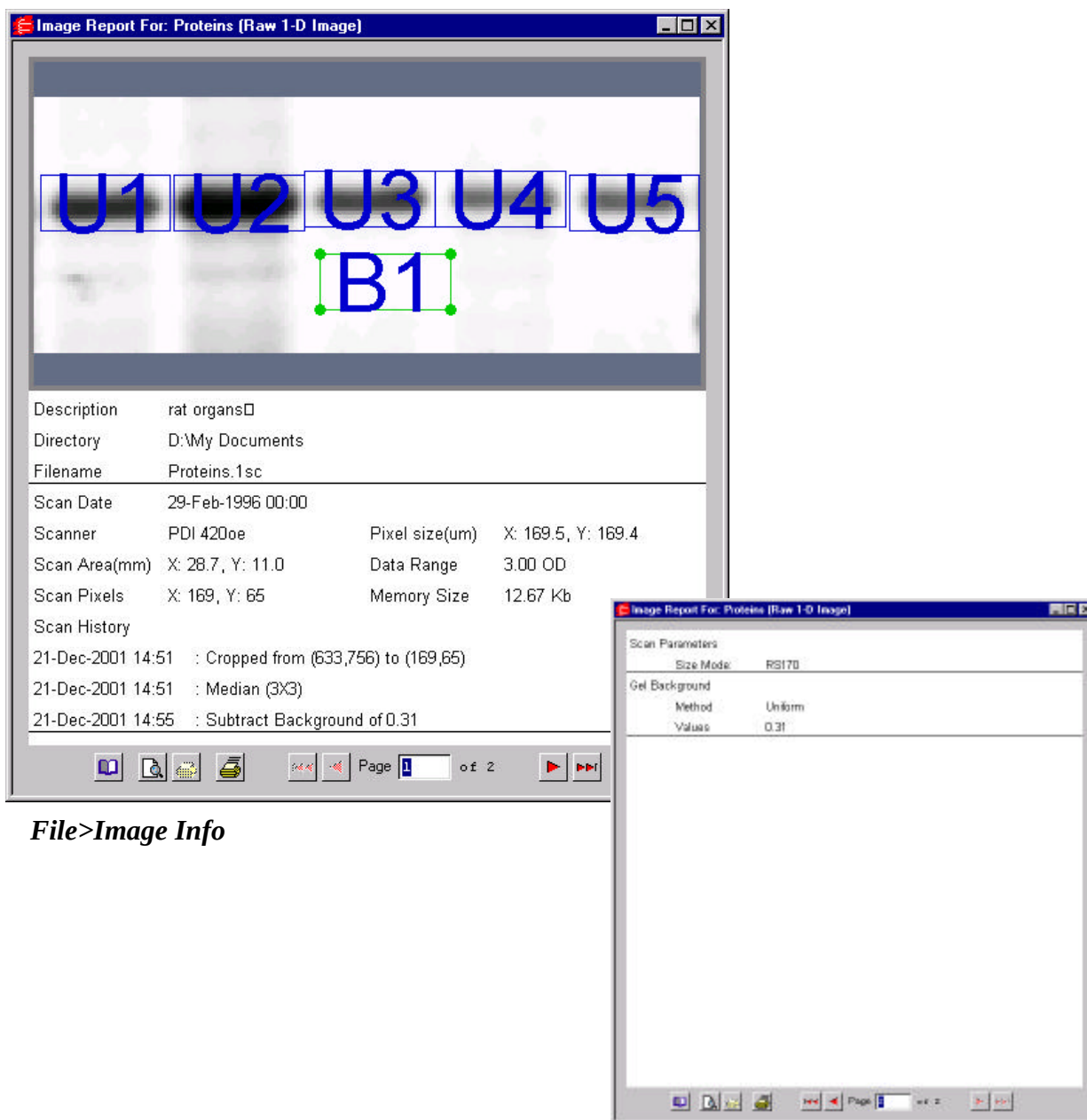
To further improve the speed of using Quantity One software and to improve efficiency by minimizing the number of mouse clicks and mouse movements, a convenient right-mouse-click menu has been added. Right-click anywhere on an open image to display a menu of common commands.



Right-click anywhere on an open image

Standard report generator for selected reports

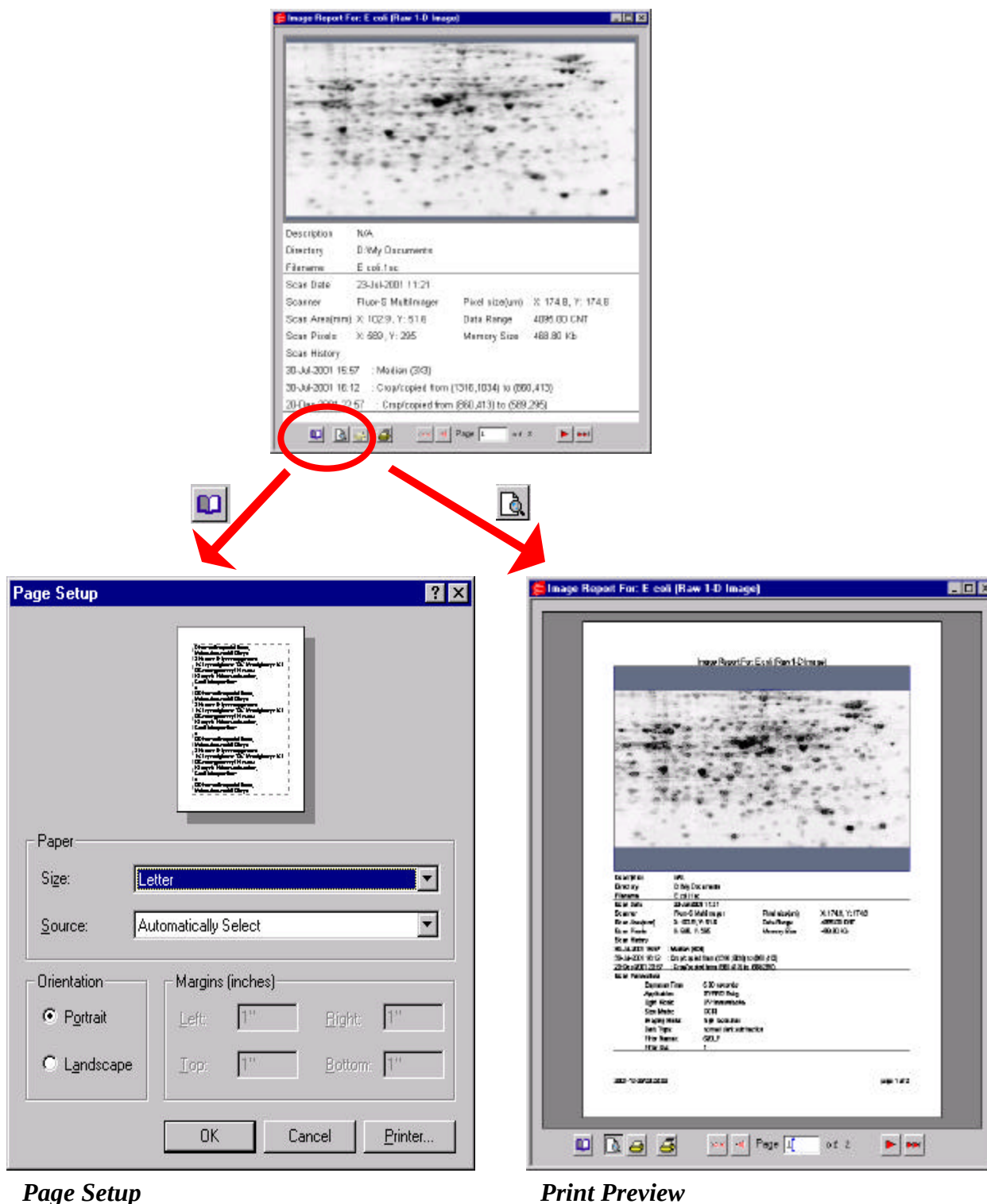
The report generator has been improved to include Page Setup and Print Preview functions. In the release of version 4.3.0, this function is available for selected reports such as the Image Report and the Band Type Presence/Absence Report. It is expected that future releases of the software will offer the improvements for other reports.



File>Image Info

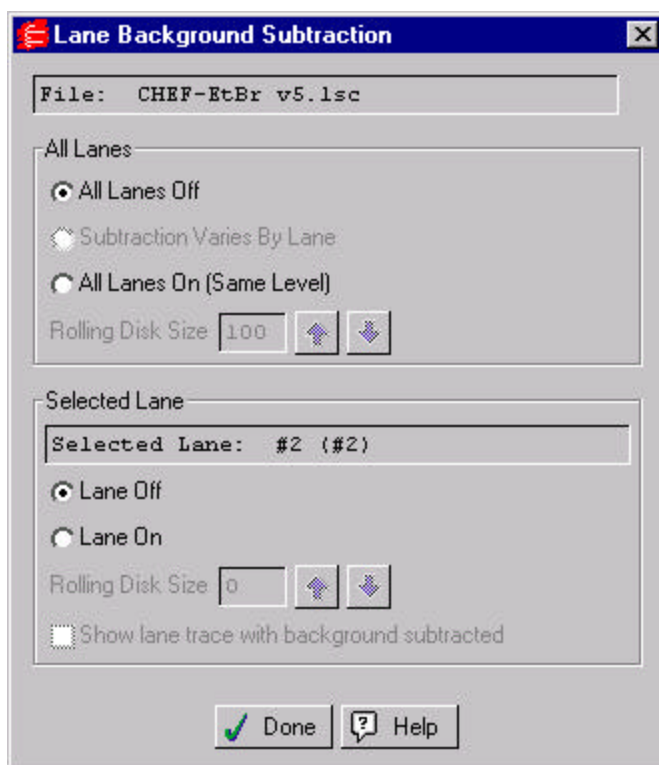
Page 2 of 2

With Image Report Page Setup, it is now convenient to customize the report and select paper orientation. The Image Report Print Preview function is particularly useful as it gives users access to information that was only obtained in a full printed report in the prior versions. It is now much more convenient to refer to information such as acquisition instrument configuration parameters as well as image alteration history, such as background subtraction methods and image filtering.



Lane background subtraction dialog improved

The Lane background dialog layout has been redesigned to be easier to work with. The functionality is the same as in previous releases of Quantity One.

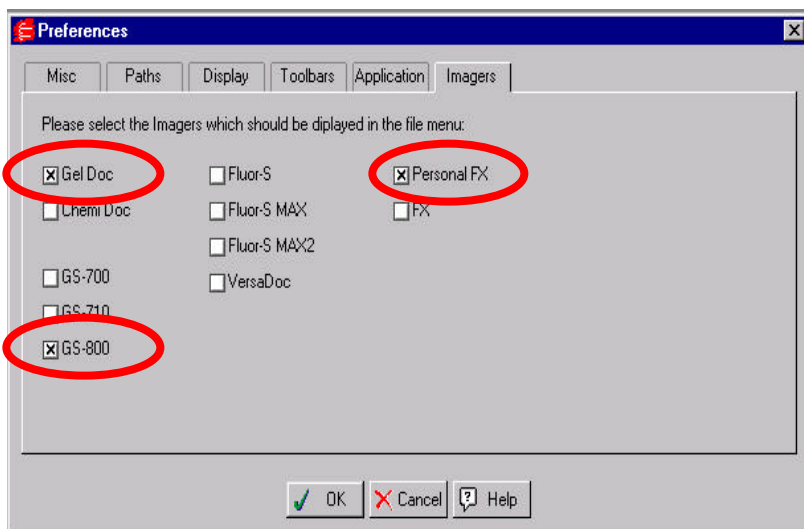


Lane>Lane Background

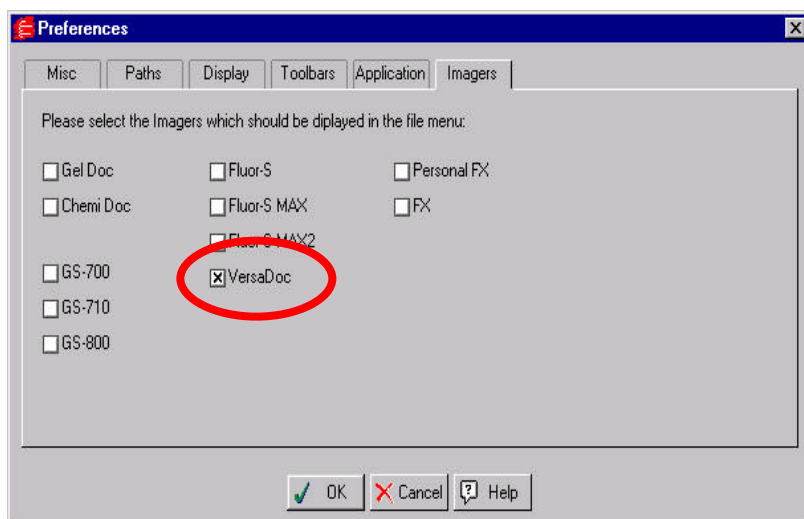
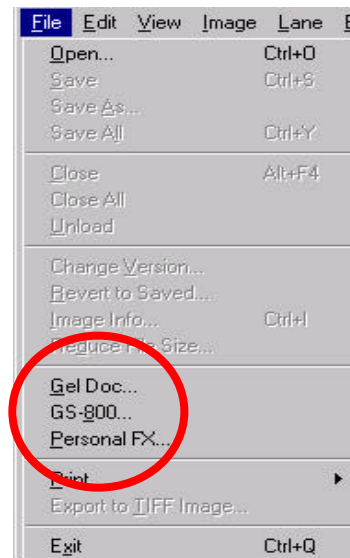
Tip! If you want to subtract individual backgrounds for the lanes, click first on the lane, then on "Lane On" in the bottom part of the dialog. "Subtraction Varies By Lane" will subsequently be activated.

Choice of which imager will be displayed in the file menu

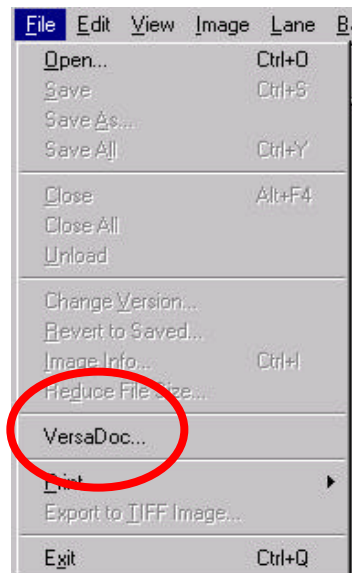
By default, all Bio-Rad imagers will be listed in the file menu. There is now an option to remove imagers that are not in use from this menu. The selection is found Edit>Preferences>Imagers.



Edit>Preferences>Imagers



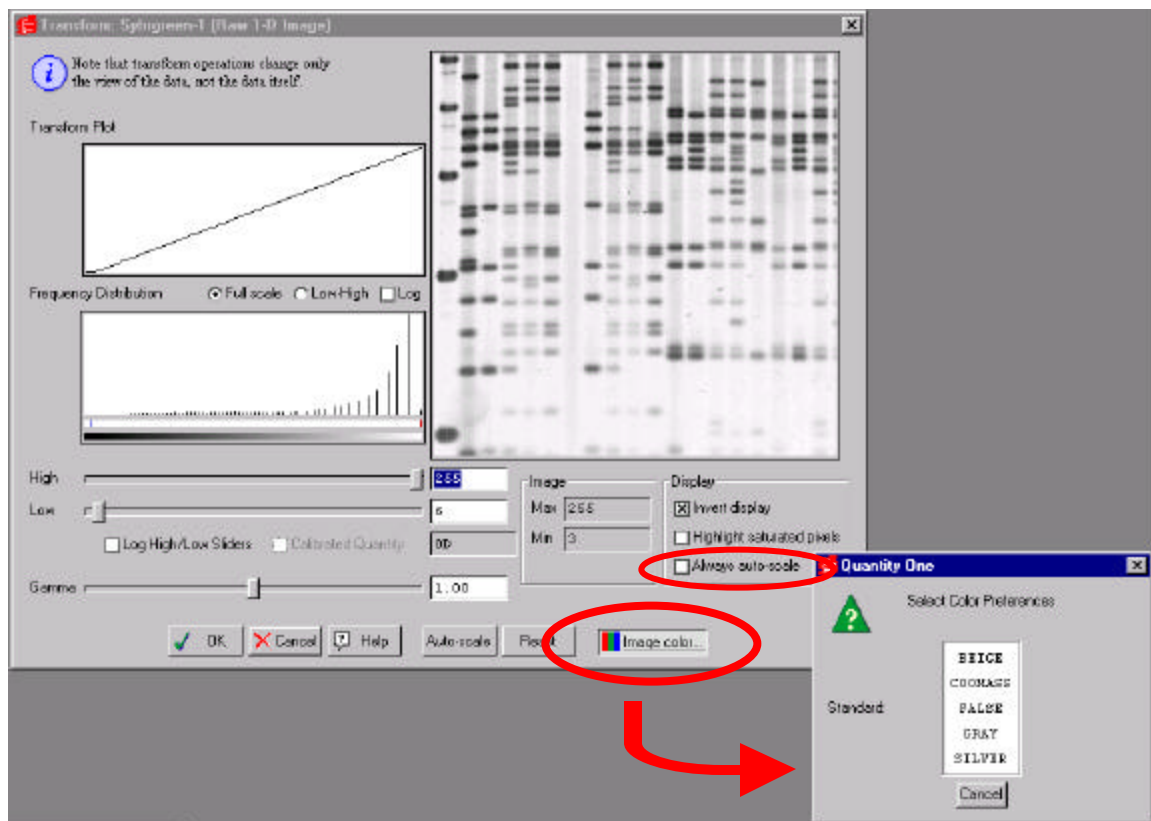
Edit>Preferences>Imagers



Transform window has two added functions

In order to offer greater flexibility in viewing images, two new optional functions have been added to the Transform window. The “always auto-scale” button allows the image window transformation to be automatically updated. The image contrast will now be auto-scaled to reflect the window view determined by the zoom factor.

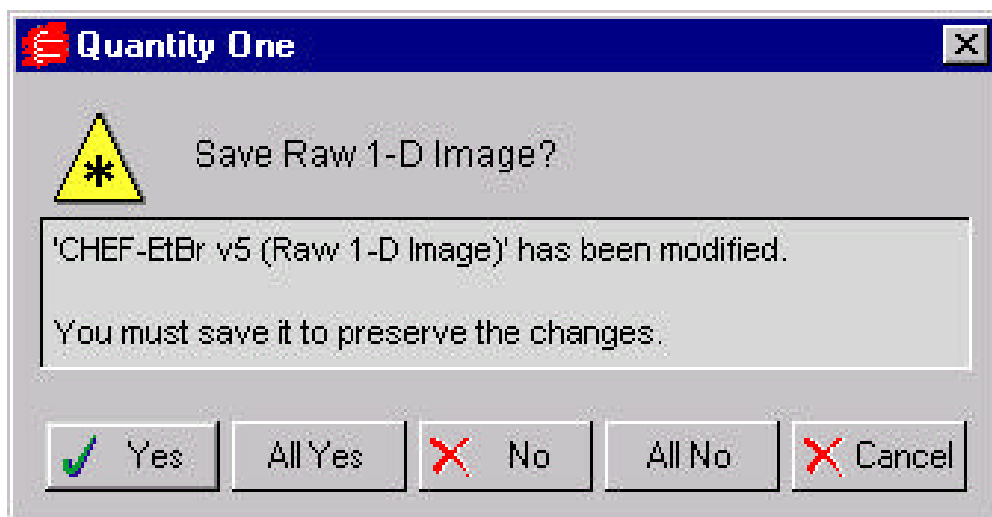
The Image Colors button accesses the Select Color Preferences dialog, which also is available from Edit>Colors>Load. This allows users to more conveniently select image color scheme preferences.



Image>Transform

Close all function added (only new for Windows)

When Edit>Close All is selected, all open images will be closed at the same time. The user is prompted to save or discard any changes made before closing.



Edit>Close All

New file system navigation (only new for Macintosh)

By introducing native file system navigation, Quantity One takes advantage of the standard directory and file browsing methods available for Windows and Macintosh. The file open dialog will default to the last used directory location.

Text Tool remembers the last font, color, size and orientation used

The Text Tool will now remember the last font, color, size and orientation used. If the settings are changed while working in one image, the Text Tool will come up with these same new settings selected when a new image is opened.

Version history

Bio-Rad Laboratories has a long history of scientific imaging and analysis software. The Discovery Series family, including Quantity One, has been used by scientists around the world to analyze and quantitate DNA/RNA and proteins since 1985. The modern Windows version of Quantity One was introduced by Bio-Rad in July 1998. Listed below are the major changes for each Bio-Rad release of Quantity One. In addition to adding drivers for new Bio-Rad Imaging Devices, enhanced functionality is added in each release.

Version	Major changes	Release	Date
4.0	Acquire images from Bio-Rad instruments Run on a PC as a Windows application (32bit) License password may be obtained via the Internet 8-bit TIFF export Auto lane finding	4.0.1	July 1998
		4.0.2	January 1999
		4.0.3	April 1999
4.1	Array Tools Colony Counting Batch Mode Free Hand Draw Volume Tool Auto-contour Volume Tools Save Molecular Weight Standards Gaussian Modeling	4.1.0	September 1999
		4.1.1	January 2000
4.2	Chemi Doc Live Acquire FX Multi-channel Sequential Scanning Multi-channel Viewer Tool Image Stack Tool Multiple Regression Models for MW Calculation Local Background Subtraction Recallable Report Templates	4.2.0	July 2000
		4.2.1	January 2001
		4.2.2	April 2001
		4.2.3	July 2001
4.3	Export image in TIFF format including overlays Band Type Presence/Absence Report New file system navigation Native printing methods for Windows/Macintosh Standard report generator for selected reports Convenient right-mouse-click menu Text Tool remembers the last font, color, size and orientation used	4.3.0	January 2002

Ordering information

Catalog number	Description
170-8601	Quantity One Software, PC
170-8609	Quantity One Software, Mac
170-8625	Quantity One Software upgrade, PC, Mac (for existing users)
170-8605	The Discovery Series CD-ROM
170-8624	Quantity One User Guide

Network licenses

The Discovery Series Software is also available as a network installation. In a laboratory with multiple software users, it is convenient to install a “floating” license of x number of copies. Please contact your local Bio-Rad office for a quotation.

Upgrading from an older version of Quantity One?

Typically, the first minor release after your purchase is free of charge. If you purchased version 4.1, you are entitled to version 4.2 and its subminor releases up to 4.2.9, free of charge.

When ordering 170-8625, Quantity One Software upgrade, PC, Mac, you will need to contact Bio-Rad Software Registration in order to receive a new password. The software, as well as the User Guide, may be downloaded from the Internet (www.bio-rad.com). You may also order the CD-ROM and the manual for a small fee.

Extended warranty

Bio-Rad offers an extended warranty program that entitles the user to receive all upgrades for a one-year period, free of charge. Please contact your local Bio-Rad office for a quotation.

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